

Intestinal Crosstalk between Bile Acids and Microbiota and Its Impact on Host Metabolism

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The gut microbiota is considered a metabolic “organ” that not only facilitates harvesting of nutrients and energy from the ingested food but also produces numerous metabolites that signal through their cognate receptors to regulate host metabolism. One such class of metabolites, bile acids, is produced in the liver from cholesterol and metabolized in the intestine by the gut microbiota. These bioconversions modulate the signaling properties of bile acids via the nuclear farnesoid X receptor and the G protein-coupled membrane receptor 5, which regulate numerous metabolic pathways in the host. Conversely, bile acids can modulate gut microbial composition both directly and indirectly through activation of innate immune genes in the small intestine. Thus, host metabolism can be affected through microbial modifications of bile acids, which lead to altered signaling via bile acid receptors, but also by altered microbiota composition.

Introduction

It is increasingly appreciated that the gut microbiota contributes to health and disease. The human gut microbiota has emerged as an environmental factor contributing to host metabolism on the basis of the initial finding that germ-free (GF) mice have reduced adiposity, a phenotype that can be reversed by colonization with a normal gut microbiota (Bäckhed et al., 2004). Obesity is via insulin resistance linked to complications such as type 2 diabetes mellitus (T2DM), cardiovascular morbidity and non-alcoholic fatty liver disease (NAFLD) and its complications liver cirrhosis and cancer. Studies in humans have demonstrated that the gut microbiota is altered in obesity and related diseases, and mouse studies have demonstrated that the gut microbiota contributes to disease phenotypes (recently reviewed in Arora and Bäckhed, 2016). However, the underlying molecular mechanisms and microbially induced signals are still at large unknown. The gut microbiota has a great potential to produce bioactive compounds that may signal to the host by activating cognate receptors in various cells (Holmes et al., 2012). One important class of microbially produced metabolites are bile acids, endogenous molecules synthesized from cholesterol in the liver, which are further metabolized by the gut microbiota (de Aguiar Vallim et al., 2013). These molecules activate receptors both in the gut, the liver and in the periphery, which can regulate several host processes, including metabolic processes.

The primary bile acids produced in humans are chenodeoxycholic acid (CDCA) and cholic acid (CA), while rodents produce CA and muricholic acids (MCAs), predominantly beta-MCA (βMCA). Before excretion into bile and further passage to the duodenum, bile acids are amidated (“conjugated”) with the amino acids glycine and to a lesser extent taurine in humans. In contrast, bile acids are almost exclusively conjugated with taurine in mice and rats (Falany et al., 1994, 1997; Vessey, 1978). Bile acids are released into the duodenum after a meal, and conjugated bile acids are then reabsorbed in the ileum and

recirculated to the liver with portal blood in a process that is called enterohepatic circulation and preserves more than 95% of the bile acid pool. de Aguiar Vallim et al. (2013) presented details of bile acid chemistry and physiology in a recent excellent review. In this perspective we summarize the current state of knowledge of how the microbiota modulates bile acids and how such regulation can contribute to metabolic disease.

Microbial Regulation of Bile Acid Synthesis

Formation of bile acids is complex and includes several reaction steps catalyzed by at least 17 different enzymes (de Aguiar Vallim et al., 2013; Russell, 2003). Bile acid synthesis takes place in the liver and can be accomplished via two different pathways (Figure 1; Table 1). The classical (or neutral) pathway accounts for at least 75% of bile acid production under normal conditions and is initiated by 7 α -hydroxylation of cholesterol catalyzed by cholesterol 7 α -hydroxylase (CYP7A1) (Thomas et al., 2008). CYP7A1 is the rate-limiting enzyme and determines the amount of bile acids produced. The alternative (or acidic) pathway is initiated by sterol-27-hydroxylase (CYP27A1) (Russell, 2003). The 27-hydroxycholesterol formed is further hydroxylated by oxysterol 7 α -hydroxylase (CYP7B1). The 7 α -hydroxylated intermediates generated from cholesterol and oxysterols then undergo sterol ring modifications and side chain oxidation and shortening in several reaction steps (Thomas et al., 2008). We have shown that the gut microbiota regulates expression of several of these enzymes, including CYP7A1, CYP7B1, and CYP27A1 (Sayin et al., 2013).

Of note, whereas the alternative pathway predominantly generates CDCA, the classical pathway generates both CDCA and CA. The ratio between these two primary bile acids is determined by the sterol 12 α -hydroxylase (CYP8B1), which is required for CA synthesis (Li-Hawkins et al., 2002). We have shown that this enzyme is not under microbial regulation (Sayin et al., 2013). In addition to CA and CDCA, mice also produce MCAs and

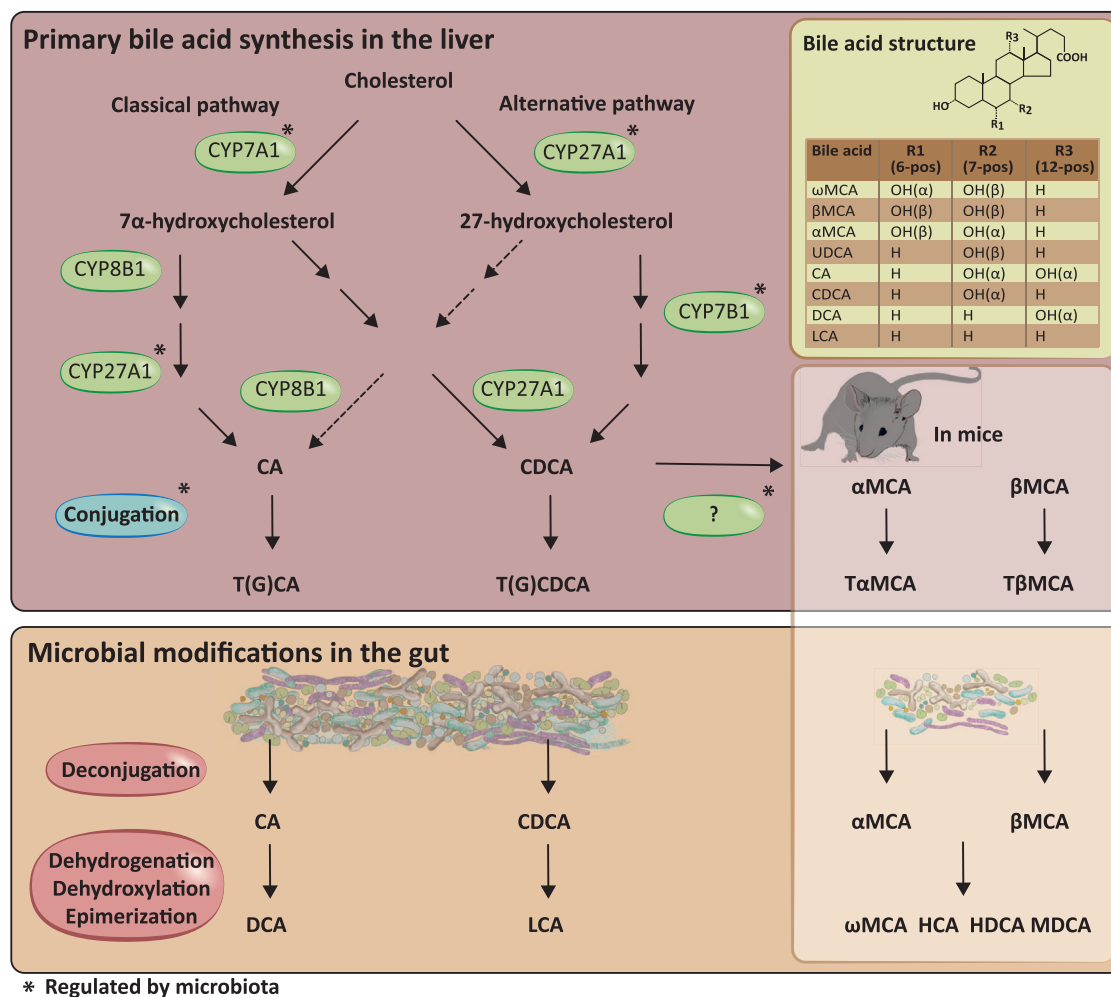


Figure 1. Bile Acid Synthesis and Metabolism

Schematic representation of synthetic pathways of primary bile acids in hepatocytes (pink) and secondary bile acids in the intestine (orange). Inset top right: table summarizing sites of hydroxylation on steroid nucleus of most common bile acid species. Inset bottom right: murine bile acid species that differ from humans. Asterisks indicate enzymes or reaction steps regulated by microbiota. G, glycine-conjugated species; T, taurine-conjugated species.

ursodeoxycholic acid (UDCA) as primary bile acids (Sayin et al., 2013). In humans, UDCA is a secondary bile acid, and MCAs are generally not detected. We and others have shown that in the absence of bacteria (as in GF or antibiotic-treated mice or rats), the bile acid pool consists of mainly primary conjugated bile acids (Kellogg et al., 1970; Kellogg and Wostmann, 1969; Koopman et al., 1986; Sayin et al., 2013; Selwyn et al., 2015; Wostmann, 1973). Early experiments in GF rats indicated that MCAs are synthesized in the liver from CDCA (Gustafsson et al., 1981); however, the enzyme(s) involved in this process are still not defined. Interestingly, the microbiota has a strong regulatory role in the synthesis of MCAs in the liver, potentially (also) regulating the unknown synthesis enzyme(s) (Sayin et al., 2013).

The liver conjugates bile acids with glycine or taurine at position C-24 in a two-step reaction (Marschall and Beuers, 2013). Synthesis of taurine as well as bile acid acyl-CoA-synthetase, which is the first of two enzymes required for bile acid conjugation, is also under microbial regulation (Sayin et al., 2013). Conjugated bile acids are then actively transported into bile via the bile salt export pump (BSEP) and may be stored in the gall-

bladder until released into the duodenum after the ingestion of a meal. The amphipathic structure of bile acids gives them detergent properties that facilitate emulsification and absorption of dietary lipids and fat-soluble vitamins. Interestingly, GF mice have increased proportion of tauro-βMCA (TβMCA) in their bile acid pool (~80%) compared with ~50% in conventionally raised (CONV-R) mice. Whether these differences also affect their capacity to emulsify lipids is unknown. Approximately 95% of biliary secreted bile acids are reabsorbed from the intestine, predominantly as conjugated bile acids in distal ileum by the apical sodium dependent bile acid transporter (ASBT, also known as IBAT), and recirculated via the portal vein to the liver, from which they are secreted again. This process is called enterohepatic circulation and occurs in humans about six times per day.

Earlier studies have shown that the ileal mucosa of GF rats has increased capacity to absorb tauro-CA (TCA), which results in a four to five times longer half-life of CA in GF compared with CONV-R rats (Gustafsson et al., 1957; Riottot and Sacquet, 1985). Together with our finding that also ASBT is under microbial regulation, this provides additional functional evidence that

Table 1. Classification of Common Bile Acid Species

Class	Metabolic Conversions	Bile Acids
Primary bile acids		
	from cholesterol by hepatic classical (neutral) or alternative (acidic) pathways involving >17 enzymes	CA, CDCA in humans CA, CDCA, UDCA, α MCA, β MCA in mice
Secondary bile acids		
	from primary bile acids	
	through gut microbial 7-dehydroxylation	DCA, LCA MDCA in mice
	from primary or secondary bile acids	
	through gut microbial 7 α / β -epimerization	UDCA in humans ω MCA in mice
	through gut microbial 3 α / β -epimerization	iso-bile acids
	through gut microbial 5 β / α -epimerization	allo-bile acids
	through gut microbial oxidation	oxo- (keto-) bile acids

the gut microbiota may not only regulate bile acid synthesis but also bile acid uptake, both contributing to the larger bile acid pool in GF mice (Sayin et al., 2013).

Microbial Metabolism of Bile Acids

Microbial deconjugation (i.e., removal of the glycine or taurine conjugate) prevents active reuptake from the small intestine via the ASBT. Bile acid deconjugation is carried out by bacteria with bile salt hydrolase (BSH) activity. Metagenomic analyses demonstrated that functional BSH is present in all major bacterial divisions and archaeal species in the human gut including members of lactobacilli, bifidobacteria, *Clostridium* and *Bacteroides* (Archer et al., 1982; Gilliland and Speck, 1977; Jones et al., 2008; Ridlon et al., 2006). In fact, BSH is enriched in the gut microbiota compared with other microbial ecosystems and is associated with increased resistance to bile toxicity (Jones et al., 2008).

Deconjugated primary bile acids that escape uptake through ASBT enter the colon, where they are metabolized through 7-dehydroxylation into secondary bile acids (Figure 1); lithocholic acid (LCA) from CDCA and deoxycholic acid (DCA) from CA (Table 1). The same reaction on the primary murine α MCA and β MCA results in the formation of murideoxycholic acid (MDCA). Omega-MCA (ω MCA) is a major metabolite of β MCA and is formed by 6 β -epimerization (Eyssen et al., 1983). Other metabolites of β MCA are hyodeoxycholic acid (HDCA), formed by 6 β -epimerization and additional 7 β -dehydroxylation, and hyocholic acid (HCA; also known as γ MCA), formed by 6 β -epimerization and further 7 β -epimerization (Degirolamo et al., 2014; Madsen et al., 1976). In humans, UDCA is formed by 7 α / β -isomerization of CDCA, which can be performed by *Clostridium absconum* (Macdonald et al., 1983; Ridlon and Bajaj, 2015). Further important isomerizations are 3 α / β -hydroxy- and 5-H β / α -epimerizations that generate iso- and allo-bile acids, respectively.

Microbial metabolism of bile acids leads to increased diversity and in general a more hydrophobic bile acid pool, which facilitates fecal elimination of bile acids, in total about 5%. A minor part of deconjugated secondary bile acids is also absorbed from the gut through passive diffusion and gets enriched in the enterohepatic circulation and may then act as signaling molecules in the host. Levels of ω MCA, MDCA, and HDCA detected in the feces of mice can vary substantially among different mouse facilities (Degirolamo et al., 2014; Sayin et al., 2013; Selwyn et al., 2015), and this could be attributed to differences in gut microbiota composition that shifts the microbial modification of the bile acids. Importantly, it has been suggested that the human gut microbiota is unable to metabolize β MCA (Martin et al., 2007, 2008; Sacquet et al., 1984, 1985). The observation that mice and humans have altered bile acid profiles that may affect signaling through bile acid receptors warrants caution when translating findings from mice to humans.

The complex process of 7-dehydroxylation comprises a number of reactions carried out by bacteria with bile acid-inducible (*bai*) genes (Doerner et al., 1997; Midtvedt, 1974; Ridlon et al., 2006). Bacteria with capability to produce secondary bile acids have been identified in *Clostridium* (clusters XIVa and XI) and in *Eubacterium*, both genera belonging to the Firmicutes phylum (Kitahara et al., 2000, 2001; Ridlon et al., 2006, 2014). Interestingly, increased levels of DCA have been associated with obesity and cancer in mice, supporting the important roles of microbial bile acid conversion for host metabolism (Yoshimoto et al., 2013).

Another major microbial biotransformation of bile acids is the generation of oxo- (or keto-) bile acids by oxidation of hydroxyl groups at ring position 3, 7, or 12 that are catalyzed by bacteria with hydroxysteroid dehydrogenases (HSDHs), which are present in Actinobacteria, Proteobacteria, Firmicutes, and Bacteroidetes (Fukiya et al., 2009; Hirano and Masuda, 1981a; Kisiela et al., 2012; MacDonald et al., 1982; Sutherland and Macdonald, 1982). These oxidation reactions are reversible and can ultimately result in epimerization. The compounds formed are commonly missed both in conventional and highly advanced bile acid analyses because of the limited availability of appropriate reference compounds. Of note, iso-bile acids are found in human serum and urine (Beuers et al., 1991) and in particular in the colon and feces (Danielsson et al., 1962), where iso-LCA and iso-DCA after LCA and DCA are the second most abundant bile acids, reaching in some individuals concentrations up to 290 and 390 μ M, respectively (Hamilton et al., 2007). Bacteria with the capability to produce iso-bile acids include *Eubacterium lentum* and *Clostridium perfringens* (Hirano and Masuda, 1981b; Hirano et al., 1981). Devlin and Fischbach (2015) showed that iso-DCA is not only produced by *E. lentum* but also by *R. garrus*, and they proposed that iso-DCA favors the growth of the abundant genus *Bacteroides* (Devlin and Fischbach, 2015), implicating microbial HSDH as a potential regulator of gut microbial composition and host metabolism. Interestingly, orally administered iso-UDCA is largely "re"-epimerized to UDCA, by the gammagamma isoform of the most abundant human liver dehydrogenase, alcohol dehydrogenase (Marschall et al., 1997, 2000). So far, little is known about 5 α -H ("allo") bile acids that are generated by dehydroxylation/elimination reactions around the original 5 β -H position (Hylemon et al.,

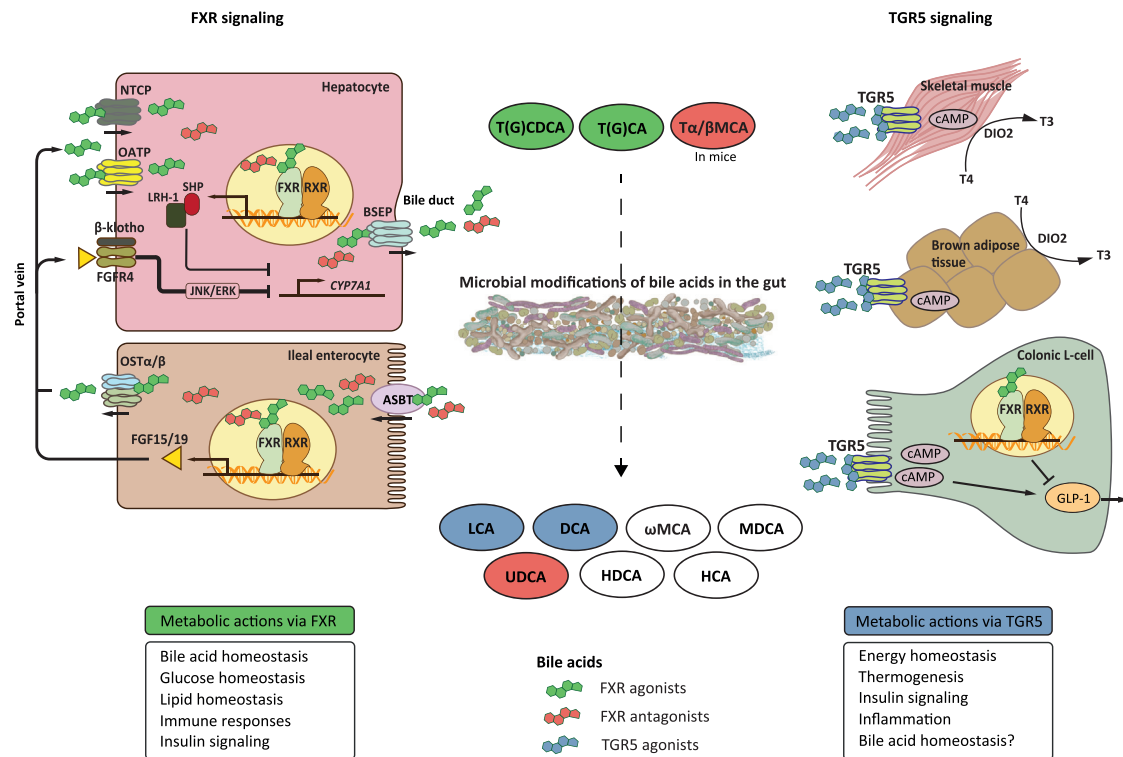


Figure 2. Microbial Modifications of Bile Acids Influence Host Metabolism via the Bile Acid Receptors FXR and TGR5

Primary bile acids T(G)CDCA, T(G)CA, and T α / β MCA (in mice) are synthesized and conjugated in the liver and transported from the hepatocytes into the bile duct by BSEP. The primary bile acids are converted into secondary bile acids by microbial modifications in the gut. The interaction between bile acids and the gut microbiota changes the bile acid composition and modulates signaling via the nuclear bile acid receptor FXR and the plasma membrane receptor TGR5. Bile acids are transported from the gut back to the liver via the enterohepatic circulation by several active transporters on ileal enterocytes (ASBT, OST α / β) and hepatocytes (NTCP and OATP).

FXR is activated mainly by the primary bile acids CDCA and CA (labeled in green), while the most potent ligands for TGR5 are LCA and DCA (labeled in blue). Other bile acids, such as T α / β MCA (primary bile acids in mice) and UDCA, have shown FXR antagonistic properties (labeled in red).

Bile acids in hepatocytes, ileal enterocytes, and colonic L cells bind to FXR and activate the FXR-RXR heterodimer complex, resulting in the transcription of target genes. In ileal enterocytes, activation of FXR leads to transcription of FGF15/19, which is secreted into the portal vein and transported to the liver, where it binds to the FGFR4/ β -klotho receptor complex on hepatocytes. The FGFR4/ β -klotho complex activates JNK/ERK signaling that inhibits expression of CYP7A1. In hepatocytes, binding of bile acids to FXR-RXR heterodimer complex results in transcription of the nuclear receptor SHP, which binds to LRH-1 and thereby also inhibits expression of CYP7A1. In colonic L-cells, FXR activation inhibits the synthesis of GLP-1.

Ligand binding of TGR5, which is a transmembrane receptor, leads to increased levels of intracellular cyclic AMP (cAMP), and this triggers further downstream signaling events. TGR5 activation in colonic L cells increases synthesis and release of GLP-1. TGR5 in skeletal muscle and brown adipose tissue increases energy expenditure by promoting the conversion of inactive thyroxine (T4) into active thyroid hormone (T3). FXR and TGR5 signaling affects many different metabolic processes in the host, and by targeting the interplay between bile acids and the gut microbiota, these processes can be altered.

DIO2, deiodinase 2; ERK, extracellular signal-regulated kinase; FGF15/19, fibroblast growth factor 15/19; FGFR4, fibroblast growth factor receptor 4; FXR, farnesoid X receptor; G, glycine-conjugated species; GLP-1, glucagon-like peptide-1; JNK, c-Jun N-terminal kinase; NTCP, sodium taurocholate cotransporting polypeptide; OATP, organic anion-transporting polypeptide; OST α / β , organic solute transporter alpha/beta; RXR, retinoid X receptor; T, taurine-conjugated species; T3, thyroid hormone; T4, thyroxine; TGR5, G protein-coupled membrane receptor 5.

1991). Allo-bile acids have a flat structure, with the A and B rings of the steroid nucleus on the same plane (A/B *trans* ring juncture) in contrast to regular bile acids in which the A and B rings are almost perpendicular (A/B *cis* ring juncture) (Hofmann and Hagey, 2008). Allo-bile acids are uncommon in healthy adults but are present in mammal fetuses and can reoccur during hepatocarcinogenesis (El-Mir et al., 2001). Further identification of the role of “exotic” bile acids and elucidation whether they signal through bile acid or other receptors may open up new avenues for how microbiota modulates signaling in the host.

Regulation of Bile Acid Synthesis via FXR: Impact of Microbiota

The synthesis of bile acids is tightly regulated by negative feedback inhibition through the nuclear receptor FXR (Sinal et al., 2000), which is a transcription factor that binds to the promoter

region and initiates the expression of a wide range of target genes (Teodoro et al., 2011). FXR is expressed in several tissues. The best-studied tissues and those with the highest FXR expression are liver and ileum (Lefebvre et al., 2009). FXR is also expressed at rather high levels in kidney and at lower levels in heart, ovary, thymus, eye, spleen, and testes (Lefebvre et al., 2009; Teodoro et al., 2011). In the liver, bile acid-activated FXR induces the expression of small heterodimer partner (SHP), which binds to liver receptor homolog-1 (LRH-1) and thereby inhibits expression of the *Cyp7a1* gene (Figure 2) (Goodwin et al., 2000; Lu et al., 2000; Sinal et al., 2000). In addition to the local effect in the liver, FXR is also activated by bile acids in the distal ileum, where it induces the expression of *Fgf15* (FGF19 in humans). FGF15/19 reaches the liver with portal blood, where it binds to the FGF receptor 4 (FGFR4)/ β -klotho heterodimer complex and triggers a JNK1/2 and ERK1/2 signaling cascade that also inhibits the

expression of *Cyp7a1* (Inagaki et al., 2005; Potthoff et al., 2012; Figure 2). Studies in tissue-specific *Fxr*-deficient mice have provided some evidence that ileal FXR activation leads to stronger suppression of *Cyp7a1*-mediated bile acid synthesis than hepatic FXR activation, whereas hepatic FXR may regulate *Cyp8b1* and thus CA formation (Kim et al., 2007; Kong et al., 2012; Zhu et al., 2011).

The most potent ligand for FXR is CDCA, followed by CA, DCA, and LCA (Makishima et al., 1999; Parks et al., 1999; Wang et al., 1999). UDCA does not activate FXR (Wang et al., 1999); it rather inhibits FXR activation (Mueller et al., 2015), which is in agreement with in silico FXR binding studies (Downes et al., 2003; Gonzalez et al., 2015). Recently, the murine taurine-conjugated primary bile acids T α MCA and T β MCA have been identified as naturally occurring FXR antagonists (Sayin et al., 2013). GF mice show accumulation of T β MCA and reduced FXR signaling and consequently increased bile acid synthesis (Sayin et al., 2013). T β MCA cannot be metabolized in the absence of bacteria. We have shown that the presence of a gut microbiota reduces the levels of T β MCA, resulting in increased expression of *Fgf15* through FXR activation, mainly in the ileum. Thus, we demonstrated that the microbiota regulates both the metabolism and the synthesis of bile acids in an FXR-dependent manner (Sayin et al., 2013). In contrast, we did not observe microbial regulation of *Cyp8b1* and CA biosynthesis.

The Gonzalez group confirmed our findings that the microbiota modulates FXR signaling. They used the antioxidant Tempol (4-hydroxy-2, 2, 6, 6-tetramethylpiperidine 1-oxyl) to alter the microbiota and bile acid profiles, which resulted in increased levels of T β MCA and suppressed FXR signaling (Li et al., 2013). Specifically, they observed that Tempol decreased the genera *Lactobacillus* and *Clostridium* (clusters IV and XIV), which was accompanied by decreased BSH activity and accumulation of intestinal T β MCA. Recently, they described the glycine conjugate of β MCA (G β MCA) also to be an FXR antagonist that in contrast to T β MCA was found to be resistant to fecal microbial BSH (Jiang et al., 2015b). However, at present it is not known whether T β MCA or G β MCA can antagonize FXR signaling in humans and how the human microbiota, which is more adapted to glycine conjugated bile acids, metabolizes G β MCA.

Microbial Regulation of Host Metabolic Processes through FXR

The importance of bile acids as signaling molecules and expression of FXR in metabolically active tissues, such as liver and small intestine, have generated vast interest in dissecting the role of FXR in metabolic disease. However, experiments with *Fxr*-deficient mice generated in part conflicting results, and it became evident that diet, and probably also the differences in microbiota among animal facilities, may contribute to these different phenotypes. *Fxr*-deficient mice on chow diet develop hyperglycemia and hypercholesterolemia (Lambert et al., 2003; Ma et al., 2006), whereas *Fxr*-deficient mice on an *Ldlr*^{-/-} background fed a high-fat diet have improved lipid profile and are protected against diet-induced obesity and atherosclerosis development (Zhang et al., 2006). In contrast, *Fxr*-deficient mice on an *Apoe*^{-/-} genetic background developed increased atherosclerosis (Hanniman et al., 2005). Nevertheless, in general it appears that *Fxr*-deficient mice fed a high-fat diet or bred on

a genetically obese background (*ob/ob*) are protected against obesity and exhibit improved glucose homeostasis compared with control mice (Parseus et al., 2016; Prawitt et al., 2011; Zhang et al., 2012). Still, the interaction between FXR and specific macronutrients warrants further investigation.

Gut microbial ecology is dramatically affected by diet in both mice and human (David et al., 2014; Turnbaugh et al., 2009; Wu et al., 2011), and it was recently demonstrated that gut microbiota from two specific breeders promoted or protected against diet-induced obesity in mice of the same genetic background (Ussar et al., 2015). We and others have demonstrated that the microbiota not only metabolizes bile acids but also affects signaling through FXR. The microbiota deconjugates the natural occurring FXR antagonist T β MCA and thus promotes FXR signaling in mice (Sayin et al., 2013) and is also required for production of secondary bile acids acting as ligands for TGR5 (Kuipers et al., 2014). Accordingly, studies using GF and antibiotic treated mice have demonstrated that the microbiota induces diet-induced obesity through FXR signaling (Jiang et al., 2015a; Li et al., 2013; Parseus et al., 2016). We found that the microbiota induced adipose tissue inflammation and increased hepatic expression of genes involved in lipid uptake in an FXR-dependent fashion (Parseus et al., 2016). Besides microbial regulation of signaling pathways, we also noted that the altered gut microbiota from high-fat fed *Fxr*-deficient mice caused less weight gain than microbiota from wild-type (WT) counterparts when transferred into GF mice (Parseus et al., 2016). This suggests that FXR-dependent microbial signaling can lead to impaired metabolism through several pathways. It is currently not known whether FXR modulation in humans leads to an altered microbiota.

Studies using intestinal and hepatic disruption of the *Fxr* have revealed that both the liver and the intestine are central organs for FXR signaling-dependent glucose and lipid control. Schmitt et al. (2015) observed that hepatic expression of FXR protects against hepatic steatosis and elevated triglyceride and bile acid levels. In contrast, mice with intestinal-specific *Fxr* deletion were protected against liver steatosis and obesity when fed a high-fat diet (Jiang et al., 2015a; Li et al., 2013). Thus, intestinal and hepatic FXR may have opposing effects on steatosis. In addition, specific intestinal FXR activation with non-absorbable fexaramine protected against the development of obesity, which was associated with enhanced thermogenesis and increased browning of adipose tissue (Fang et al., 2015). Interestingly, this treatment increased levels of LCA, which suggests that some of these effects might be related to an altered microbiota and mediated via TGR5, which is known to activate thermogenesis (Watanabe et al., 2006). Thus, a picture emerges that FXR may have beneficial effects on chow diet, whereas on high-fat diet, *Fxr*-deficiency may protect against metabolic disease, in particular, when intestinal FXR is inhibited. It will be essential to clarify how the gut microbiota affects FXR signaling in the small intestine and liver in humans.

Bile Acids and FXR Modify the Microbiota in a Diet-Dependent Fashion

The interaction between the microbiota and bile acids is not unidirectional. Bile acids can shape the gut microbiota community by promoting the growth of bile acid-metabolizing bacteria and inhibiting the growth of other bile sensitive bacteria. Studies

have shown that biliary obstruction, which blocks the bile flow into the gut, leads to bacterial overgrowth and translocation of bacteria in the small intestine (Clements et al., 1996), a phenotype that can be reversed with bile acid administration (Lorenzo-Zúñiga et al., 2003). It became evident that bile acids not only have direct antimicrobial effects, by destroying bacterial membranes due to their detergent properties, but also have indirect effects through FXR, by inducing transcription of antimicrobial agents (e.g., iNOS and IL-18) that affect the gut microbiota via the immune system (Inagaki et al., 2006).

We have recently shown that FXR can modulate the microbiota and that these alterations were associated with changes in the bile acid composition (Parseus et al., 2016). *Fxr*-deficient mice on a high-fat diet had increased relative abundance of Firmicutes and decreased abundance of Bacteroidetes compared with WT mice on the same diet. Bile acid profiles in serum and caecum were also altered with higher levels of T β MCA and β MCA, which could be due to less conversion of these primary bile acids into secondary bile acids. Accumulation of primary bile acids in the *Fxr*-deficient mice indicates that the microbiota might have reduced bile acid metabolizing capacity in the absence of FXR. GF *Fxr*-deficient mice showed similar bile acid profiles as their WT counterparts, demonstrating that the FXR-mediated effect on bile acid composition is dependent on the altered microbiota.

Devkota et al. (2012) further showed strong interactions between diet, bile acids and the microbiota. Mice fed with a high-fat diet composed of either milk fat or polyunsaturated fat were compared with mice on chow diet. High-fat diet increased the abundance of Bacteroidetes and decreased abundance of Firmicutes regardless of the source of fat. Interestingly, the milk-fat diet led to a shift in the bile acid composition with increased amounts of TCA and an expansion of *Bilophila wadsworthia* that was not found in the mice fed polyunsaturated fat (Devkota et al., 2012). *B. wadsworthia* is a member of the delta Proteobacteria and is recognized as a “bile-loving” bacterium because of its requirement for bile in the growth medium (Baron et al., 1989). These types of bacteria have been associated with inflammatory bowel diseases, and indeed, a correlation between *B. wadsworthia* and the severity of colitis in IL-10-deficient mice could be demonstrated (Devkota et al., 2012). An increase in *B. wadsworthia* was observed when *IL-10*^{-/-} mice on chow diet were gavaged with TCA for a week, and almost the same bloom of *B. wadsworthia* was observed as in those mice on milk-fat diet. Furthermore, supplementation of the diet with omega-3 fish oil is supposed to inhibit the bloom of *B. wadsworthia*, most likely because of alterations in the bile acid composition (Devkota and Chang, 2015). This is in agreement with a recent study demonstrating that the abundance of *Bilophila* is reduced after feeding mice with fish oil rather than lard as fat source (Caesar et al., 2015). These studies show that there is a dynamic interplay between bile acids and microbiota that can be modified by diet causing beneficial or detrimental effects on host metabolism.

It is essential to understand the interplay among the microbiota, diet, and bile acids to clarify how the microbiota modulate metabolic outcome. An important example is the widely accepted microbial metabolism of choline and carnitine that produces trimethylamines, which are oxidized by flavin mono-ox-

genase 3 (FMO3) to generate proatherogenic trimethylamine-N-oxide (TMAO) (Aron-Wisniewsky and Clément, 2016; Koeth et al., 2013; Wang et al., 2011). Interestingly, it has been shown that FMO3 is under the regulation of FXR. Bennette et al. (2013) demonstrated that WT but not *Fxr*-deficient mice treated with natural (CA) or synthetic (GSK2324 or GW4064) FXR ligands exhibited induction of FMO3 expression and increased TMAO levels. Furthermore, they showed that hepatic FMO3 expression is regulated by hepatic, not intestinal, FXR (Bennett et al., 2013; Koeth et al., 2013). In addition, Koeth et al. (2013) showed that a diet supplemented with carnitine or TMAO reduced the bile acid pool in mice, which could affect FXR signaling and alter metabolic phenotypes. Thus, interplay among microbiota, bile acids, FXR, and co-metabolites from microbial and host metabolism may contribute to metabolic disease.

The Role of TGR5 in Host Metabolism

TGR5 is another bile acid-responsive receptor involved in host metabolism. TGR5 is a plasma membrane-bound G protein-coupled receptor ubiquitously expressed with high expression in gallbladder, placenta, lung, spleen, intestine, liver, brown and white adipose tissue, skeletal muscle, and bone marrow (Kawamata et al., 2003; Maruyama et al., 2002, 2006; Vassileva et al., 2006). TGR5 is activated mainly by the secondary bile acids LCA and DCA (Chen et al., 2011; Maruyama et al., 2002) and is therefore an interesting target in the context of microbiota-bile acid interactions (Figure 2).

Tgr5-deficient mice were generated in 2006, and they appeared healthy and fertile, with no obvious abnormalities. Interestingly, they had a decreased bile acid pool, suggesting a role for TGR5 in bile acid homeostasis (Maruyama et al., 2006; Vassileva et al., 2006), but it is unclear whether this is a direct effect of TGR5 or mediated through FXR. In addition, *Tgr5*-deficient mice were protected against cholesterol gallstone formation when fed a lithogenic diet (Vassileva et al., 2006).

Watanabe et al. (2006) showed that TGR5 might have a role in energy homeostasis by promoting intracellular thyroid hormone activity and thereby increasing thermogenesis in brown adipose tissue. They found that high-fat diet-induced obesity and fat accumulation in brown adipose tissue was completely reversed by dietary CA supplementation, suggesting an FXR-dependent mechanism, but this was excluded by the use of the highly active synthetic FXR ligand GW4064. Further analyses revealed that the weak TGR5 agonist CA was metabolized to the stronger TGR5 agonist DCA (Watanabe et al., 2006). Another study confirmed that TGR5 signaling controls glucose homeostasis by increased energy expenditure in brown adipose tissue and muscle and by increased GLP-1 release in intestinal L cells (Thomas et al., 2009; Figure 2). Interestingly, we have recently shown that L cells also express FXR and that FXR also regulates GLP-1 synthesis (Trabelsi et al., 2015). In high-fat diet-induced obese mice, the TGR5-specific agonist INT-777, which is a derivative of CDCA, ameliorated hepatic steatosis and adiposity and improved insulin sensitivity (Thomas et al., 2009). At present it is unclear whether FXR and TGR5 signal directly or indirectly via alterations in the microbiota and bile acid metabolism that produce agonistic or antagonistic signaling through the other receptor. One such recent example was demonstrated when healthy human subjects were treated with CDCA for 2 days and

showed increased brown adipose tissue activity and whole-body energy expenditure. In accompanying *in vitro* experiments, it was shown that the FXR agonist GW4064 was ineffective while LCA and the TGR5 agonists CpdA and INT-777 had similar effects as CDCA (Broeders et al., 2015). These findings strongly indicate that when providing natural bile acids that are agonists for FXR the gut microbial metabolism may generate ligands for TGR5, emphasizing the importance of studying the gut microbiota.

Modulations of Microbiota-Bile Acid Interactions in NAFLD

NAFLD is the hepatic manifestation of the metabolic syndrome and results from dysregulated lipid and glucose metabolism (Carr and Reid, 2015). NAFLD is a complex disease spanning from simple steatosis to non-alcoholic steatohepatitis (NASH) with fibrosis, cirrhosis, and ultimately liver cancer. Accumulating data suggest that the microbiota has a role in the development of NAFLD (Moschen et al., 2013; Schnabl and Brenner, 2014). Patients with NAFLD have lower expression of FXR and increased levels of triglycerides in serum (Yang et al., 2010).

There is currently no treatment for patients with NAFLD/NASH other than dietary and lifestyle recommendations, but bile acid receptor activation is among the most intensively debated novel treatment options (Schaap et al., 2014), including the semisynthetic bile acid derivative obeticholic acid (OCA; 6 α -ethyl-CDCA, INT-747). OCA is a 100 times more potent ligand to FXR than CDCA and has been tested in phase 2 and 3 trials in patients with NASH, with and without T2DM. Treatment with OCA for 6 weeks in patients with NAFLD and T2DM dose-dependently improved insulin sensitivity, estimated by euglycemic clamp, and reduced body weight, compared with placebo (Mudaliar et al., 2013). A multicenter, double-blind, randomized, placebo-controlled, phase 3a study with OCA (FLINT) in 283 patients with NASH, with and without T2DM, demonstrated that 72 weeks of treatment with OCA led to significant improvement in the NAS (NASH activity score). Improvement of fibrosis was also observed in the OCA treatment group compared with controls (Neuschwander-Tetri et al., 2015). Our ongoing investigator-initiated trials with OCA aim to decipher the molecular actions of FXR activation in humans with NAFLD and the impact of OCA on the gut microbiota.

Changes in Microbiota-Bile Acid Interactions after Bariatric Surgery

Bariatric surgery is at present the most effective procedure for the treatment of obesity. We recently demonstrated that bariatric surgery has long-term effects on the gut microbiota composition and function (Tremaroli et al., 2015). The altered microbiota from bariatric surgery patients transplanted into GF mice reduced body fat gain compared with microbiota from obese control patients without surgery. Furthermore, Roux-en-Y gastric bypass (RYGB) patients showed increased postprandial bile acid and FGF19 response compared with obese controls, which is in line with a previous study (Jansen et al., 2011). We have also shown that FXR is an important contributor to the weight loss and improved glucose metabolism after bariatric surgery in mice (Ryan et al., 2014). In contrast, experiments with *Tgr5*-deficient mice showed that TGR5 did not contribute to the reduced body weight, increased energy expenditure, or

increased glucose-stimulated insulin secretion after vertical sleeve gastrectomy (VSG) (McGavigan et al., 2015). However, glucose tolerance, hepatic insulin signaling, and metabolically favorable changes in bile acid profiles after VSG were attenuated in the *Tgr5*-deficient mice, indicating that TGR5 has an impact on at least some of the beneficial effects after bariatric surgery.

A recent study by Flynn et al. (2015) showed similar positive effects in terms of weight loss and improved glucose tolerance after bile diversion (with reconstructed bile acid-release into the ileum) as after RYGB surgery in mice. A significant increase in serum T β MCA and T ω MCA levels and, conversely, reduced FXR signaling were observed after bile diversion, in contrast to our previous finding of induced FXR after VSG in mice (Ryan et al., 2014). This underlines the complexity of the role of FXR as a regulator of host metabolism where beneficial effects can be achieved both by reducing and by increasing FXR signaling in a context-specific manner. Furthermore, it remains to be demonstrated how intestinal FXR signaling is modulated in humans after bariatric surgery.

Conclusions

The bile acid receptors FXR and TGR5 have become major targets for translational and interventional studies of metabolic diseases. It is clear that the microbiota can modulate signaling through both FXR and TGR5 via modifications of bile acids. Thus understanding of the gut microbiota and its implication on bile acid metabolism and signaling may provide understanding for how pre-clinical data can be interpreted. However, human and murine bile acids have substantially different signaling properties, and less abundant and/or characterized bile acid metabolites may additionally hamper proper translational conclusions. Differences in the microbiota may also explain an individual's response to bile acid-derived drugs, and deeper understanding of the microbiota may identify individuals that will respond beneficially to a treatment and those who will experience adverse effects. Thus, targeting the interplay between microbiota, bile acids, and FXR and/or TGR5 signaling seems to evolve as a promising avenue for the treatment of metabolic diseases, but much more research is needed.

AUTHOR CONTRIBUTIONS

All authors wrote the paper and approved the final version.

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